



Task 04: Security Operations Center (SOC)- Practical Evidence Collection and Threat Intelligence Validation

Cybersecurity Internship – CyArt

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Introduction

This task focuses on practical implementation of Security Operations Center (SOC) activities using real-world security tools and simulated screenshots for evidence-based learning. The objective of this task is to understand how SOC analysts investigate security incidents using threat intelligence platforms, SIEM dashboards, log analysis tools, and forensic validation systems.

In a real SOC environment, analysts rely heavily on visual evidence such as dashboards, alert screens, and threat intelligence reports to identify and validate cyber threats. These screenshots represent how security events are monitored, analyzed, and correlated across multiple platforms.

This task includes practical demonstration of threat intelligence validation, alert investigation, and incident analysis using tools such as AlienVault OTX, SIEM dashboards, and security monitoring systems. Each screenshot represents a real SOC workflow step, helping to understand how malicious IPs, suspicious activities, and security alerts are identified and investigated.

The main aim of this task is to develop hands-on understanding of SOC operations by analyzing realistic security interfaces and learning how professionals interpret security data during incident response and threat hunting processes.



1. Threat Hunting Practice

Introduction

Threat Hunting is a proactive cybersecurity process used to search for hidden threats in a network before they trigger alerts. Unlike traditional incident response, threat hunting is hypothesis-driven and focuses on identifying attacker behavior using logs, endpoints, and threat intelligence.

It is important for:

- detecting hidden threats in systems
- identifying advanced persistent threats (APT)
- improving SOC detection capabilities
- reducing dwell time of attackers
- proactive security monitoring

The main goal is to find suspicious activity that is not detected by automated tools.

Core Concepts

1. Hypothesis-Based Hunting

Threat hunting starts with a hypothesis.

Example:

“An attacker is using valid accounts (T1078) for unauthorized access.”

2. Data Sources Used

- SIEM logs (Elastic / Wazuh)
- Endpoint logs (Velociraptor)
- Network traffic logs
- Threat intelligence feeds (OTX)

3. MITRE ATT&CK Mapping



Hunting is mapped to attacker techniques like:

- T1078 → Valid Accounts
- T1055 → Process Injection

Practical Knowledge

Practical 1: Hypothesis-Based Hunt

Hypothesis:

“Unauthorized privilege escalation is happening in domain accounts.”

Search Query:

Event ID 4672 (special privileges assigned)

Result Table:

Timestamp	User	Event ID	Notes
-----	-----	-----	-----
2025-08-18 15:00:00	testuser	4672	Unexpected admin role

Observation:

User received elevated privileges without approval → suspicious activity.

Practical 2: Threat Intelligence Hunt

Tool: AlienVault OTX

Steps:

- Search for T1078 indicators
- Check suspicious IPs and domains

Observation:

Some IPs matched known malicious activity from threat feeds.

Practical 3: Endpoint Validation



Tool: Velociraptor

Query:

```
SELECT * FROM processes
```

Observation:

- Unknown processes detected
- Possible privilege escalation activity

Practical 4: Hunting Report

Summary (100 words):

Threat hunting was performed using a hypothesis-based approach focusing on unauthorized privilege escalation (T1078). SIEM logs were analyzed to identify Event ID 4672, which showed unexpected administrative privileges assigned to a user account. Endpoint analysis using Velociraptor revealed suspicious processes running without authorization. Threat intelligence feeds from AlienVault OTX were used to validate potential malicious indicators. The findings suggest possible misuse of valid credentials for unauthorized access. Further investigation and monitoring are recommended to confirm compromise and prevent escalation.

Tools Used

- Elastic Security
- Velociraptor
- AlienVault OTX
- Wazuh
- MITRE ATT&CK

Skills Learned

- proactive threat hunting
- hypothesis creation
- log analysis



- MITRE mapping
 - endpoint investigation
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2. SOAR Playbook Development

Introduction

SOAR (Security Orchestration, Automation, and Response) is used to automate and coordinate security operations tasks in a SOC. It helps reduce manual effort by automating repetitive incident response actions such as alert validation, ticket creation, and IP blocking.

It is important for:

- automating SOC workflows
- improving incident response speed
- reducing manual workload
- standardizing security actions
- integrating SIEM and response tools

The main goal is to automate incident response using structured playbooks.

Core Concepts

1. SOAR Components

- Orchestration: connects different security tools
- Automation: performs repetitive tasks automatically
- Response: takes action on incidents (block, isolate, alert)

2. Playbook Concept

A playbook is a predefined workflow that executes actions step by step during a security incident.

Example:

Phishing email → check IP → validate threat → block IP → create ticket



3. Integration

SOAR tools integrate with:

- SIEM (Wazuh / Elastic)
- Threat intelligence (OTX / VirusTotal)
- Response tools (CrowdSec / firewall)

Practical Knowledge

Practical 1: Playbook Design

Playbook: Phishing Response Automation

Steps:

1. Detect phishing alert in Wazuh
2. Extract IP address
3. Check IP reputation (OTX / VirusTotal)
4. If malicious → block IP using CrowdSec
5. Create incident ticket in TheHive

Practical 2: Playbook Execution Test

Playbook Step	Status	Notes
Check IP	Success	IP flagged as malicious
Validate IP	Success	Confirmed via OTX
Block IP	Success	CrowdSec blocked IP
Create Ticket	Success	TheHive case generated

Observation:

Playbook successfully automated phishing incident response.

Practical 3: Automation Summary

Summary (50 words):



A SOAR playbook was developed to automate phishing incident response. The workflow included detection of malicious alerts, validation using threat intelligence, automated IP blocking via CrowdSec, and case creation in TheHive. The automation reduced manual effort and improved response time, ensuring faster mitigation of phishing threats.

Tools Used

- Splunk SOAR (Phantom)
- TheHive
- Wazuh
- CrowdSec
- VirusTotal
- AlienVault OTX

Skills Learned

- SOAR automation
- playbook creation
- security orchestration
- incident response automation
- tool integration

3. Post-Incident Analysis and Continuous Improvement

Introduction

Post-Incident Analysis is the process of reviewing and analyzing a security incident after it has been resolved. It helps identify root causes, measure response effectiveness, and improve SOC processes to prevent future incidents.

It is important for:

- identifying root cause of incidents
- improving SOC processes and response time
- analyzing security gaps
- learning from past incidents



- improving overall security posture

The main goal is to understand what happened, why it happened, and how to prevent it in the future.

Core Concepts

1. Root Cause Analysis (RCA)

RCA is used to find the main reason behind a security incident.

Methods:

- 5 Whys technique
- Fishbone diagram

2. SOC Metrics

- MTTD (Mean Time to Detect)
- MTTR (Mean Time to Respond)
- Dwell Time (time attacker stays undetected)

3. Continuous Improvement

Findings from incidents are used to:

- improve detection rules
- update security policies
- enhance SOC workflows

Practical Knowledge

Practical 1: Root Cause Analysis (5 Whys)

Incident: Phishing Attack

| Question | Answer |

|-----|-----|



Why was email opened?	User clicked malicious link
Why clicked link?	Weak email filtering
Why weak filtering?	No advanced email security
Why no security?	System not updated
Why not updated?	No patch management policy

Observation:

Main cause → weak email security controls

Practical 2: Fishbone Diagram

Categories analyzed:

- People (user awareness)
- Process (security policies)
- Technology (email filters)
- Tools (security systems)

Observation:

Multiple factors contributed to phishing success.

Practical 3: Metrics Calculation

Incident Timeline:

- Detection Time: 2 hours
- Response Time: 4 hours

Results:

- MTTD = 2 hours
- MTTR = 4 hours

Summary (50 words):

The incident showed a Mean Time to Detect of 2 hours and Mean Time to Respond of 4 hours. This indicates moderate detection speed but delayed response. Improvements in automation and alert prioritization are required to reduce response time and enhance SOC efficiency.



Tools Used

- Google Sheets
- Draw.io
- Wazuh
- Elastic Security
- SOC reporting tools

Skills Learned

- root cause analysis
 - SOC performance evaluation
 - incident review process
 - metrics calculation
 - continuous improvement strategies
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4. Alert Triage with Automation

Introduction

Alert Triage with Automation is the process of analyzing security alerts and automatically validating them using security tools and threat intelligence. It combines manual triage with automated workflows to improve speed and accuracy in SOC operations.

It is important for:

- reducing false positives
- speeding up alert analysis
- improving incident response efficiency
- validating alerts using threat intelligence
- automating repetitive SOC tasks

The main goal is to quickly classify alerts and automate their validation and response wherever possible.



Core Concepts

1. Alert Triage

Alert triage is the process of classifying alerts based on severity and relevance:

- High: confirmed threat
- Medium: suspicious activity
- Low: false positive or benign

2. Automation in Triage

Automation is used to:

- check IP reputation
- validate file hashes
- enrich alerts with threat intelligence
- trigger response actions automatically

3. Threat Intelligence Integration

Tools like VirusTotal and OTX are used to:

- verify malicious indicators
- enrich alert context
- support decision making

Practical Knowledge

Practical 1: Alert Analysis

Alert Type: Suspicious File Download

Alert ID	Description	Source IP	Priority	Status
-----	-----	-----	-----	-----
005	File Download	192.168.1.102	High	Open

Observation:

Suspicious file download activity detected from unknown IP.



Practical 2: Automated Validation

Steps:

- Extract file hash or IP
- Check in VirusTotal
- Validate using OTX
- Mark status automatically

Result:

- IP flagged as suspicious/malicious
- Alert marked for escalation

Practical 3: Automation Flow

Workflow:

1. Alert generated in Wazuh
2. TheHive receives alert
3. VirusTotal checks file/IP
4. If malicious → escalate to Tier 2
5. Case created automatically

Practical 4: Summary

Summary (50 words):

Alert triage with automation was performed using SIEM and threat intelligence tools. Suspicious file download alerts were analyzed, and automated validation was performed using VirusTotal and OTX. The system successfully classified the alert as high severity and escalated it for further investigation, improving SOC response efficiency.

Tools Used

- Wazuh



- VirusTotal
- TheHive
- AlienVault OTX
- SOAR tools
- SIEM dashboards

Skills Learned

- alert triage process
 - security automation
 - threat intelligence validation
 - incident classification
 - SOC workflow optimization
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5. Evidence Analysis

Introduction

Evidence Analysis is the process of examining collected digital evidence to identify suspicious activity, attacker behavior, and system compromise. It is a key part of digital forensics and SOC investigations, ensuring proper understanding of incidents through structured analysis.

It is important for:

- analyzing collected forensic data
- identifying attacker behavior
- validating security incidents
- supporting incident response
- maintaining investigation accuracy

The main goal is to extract meaningful insights from collected evidence without altering its integrity.

Core Concepts



1. Evidence Types

- Network logs
- Process data
- System logs
- Memory data
- Authentication logs

2. Chain of Custody

Ensures evidence integrity by tracking:

- who collected evidence
- when it was collected
- where it was stored
- who accessed it

3. Log and Network Analysis

Used to:

- detect suspicious connections
- identify malicious processes
- trace attacker activity

Practical Knowledge

Practical 1: Network Analysis

Tool: Velociraptor

Query:

```
SELECT * FROM netstat
```

Observation:

- active network connections identified
- suspicious external IP connections detected



Practical 2: Process Investigation

Command:

```
SELECT * FROM processes
```

Observation:

- unknown processes running in system
- possible malware execution detected

Practical 3: Evidence Documentation

Item	Description	Collected By	Date	Hash Value
-----	-----	-----	-----	-----
Network Log	Server-Z Log	SOC Analyst	2025-08-18	<SHA256>

Observation:

Proper documentation ensures evidence traceability and integrity.

Practical 4: File Integrity Check

Command:

```
sha256sum suspiciousfile.exe
```

Observation:

- verifies file integrity
- ensures evidence has not been modified

Practical 5: Summary

Summary (50 words):

Evidence analysis was performed using network and process investigation techniques. Suspicious connections and unknown processes were identified using Velociraptor. File integrity was verified using SHA256 hashing. Proper documentation ensured chain of custody was maintained, supporting accurate forensic investigation and ensuring evidence reliability for incident response.



Tools Used

- Velociraptor
- FTK Imager
- Linux forensic commands
- SHA256 tools
- SOC investigation tools

Skills Learned

- digital evidence analysis
- process and network investigation
- chain of custody management
- forensic documentation
- file integrity verification

6. Adversary Emulation Practice

Introduction

Adversary Emulation Practice is the process of simulating real-world attacker behaviors in a controlled environment to test SOC detection and response capabilities. It helps organizations understand how well their security systems can detect and respond to real cyber attacks.

It is important for:

- testing SOC detection capabilities
- simulating real attacker techniques
- improving security controls
- mapping attacks to MITRE ATT&CK
- strengthening incident response readiness

The main goal is to replicate attacker behavior safely and evaluate how effectively the SOC detects it.



Core Concepts

1. Adversary Emulation

It involves simulating attacker tactics such as:

- phishing attacks (T1566)
- exploitation of services (T1210)
- credential misuse (T1078)

2. Emulation Frameworks

- MITRE Caldera (automated attack simulation)
- Red team tools
- Controlled penetration testing tools

3. SOC Validation

Emulation helps validate:

- SIEM alerts
- detection rules
- incident response workflows

Practical Knowledge

Practical 1: Attack Simulation

Tool: MITRE Caldera

Scenario:

Simulate spear phishing attack (T1566)

Observation:

- phishing attempt executed in controlled environment
- logs generated in SIEM for detection



Practical 2: Detection Validation

Tool: Wazuh

Timestamp	TTP	Detection Status	Notes
----- ----- ----- -----			
2025-08-18 17:00:00	T1566	Detected	Phishing blocked

Observation:

SOC successfully detected phishing simulation.

Practical 3: SOC Response Check

Observation:

- alert generated in Wazuh
- incident logged in SIEM
- detection rules triggered correctly

Practical 4: Summary

Summary (100 words):

Adversary emulation was performed using MITRE Caldera to simulate a spear phishing attack (T1566). The attack was executed in a controlled environment to test SOC detection capabilities. Wazuh successfully detected the phishing activity and generated an alert in real time. The incident was logged and mapped to MITRE ATT&CK framework for analysis. This exercise validated the effectiveness of existing detection rules and highlighted the importance of continuous security testing. It also helped identify areas where detection rules can be improved to strengthen SOC readiness against real-world phishing attacks.

Tools Used

- MITRE Caldera
- Wazuh



- Elastic Security
- SIEM tools
- SOC monitoring systems

Skills Learned

- adversary emulation
- MITRE ATT&CK mapping
- SOC validation testing
- attack simulation
- detection rule evaluation

7. Security Metrics and Executive Reporting

Introduction

Security Metrics and Executive Reporting focuses on measuring SOC performance and presenting security information in a clear and understandable way for technical and non-technical stakeholders. It helps organizations track efficiency, detect gaps, and improve overall cybersecurity posture.

It is important for:

- measuring SOC performance
- tracking incident response efficiency
- identifying security gaps
- supporting decision-making for management
- improving communication with executives

The main goal is to analyze SOC performance using metrics and present findings in an executive-friendly format.

Core Concepts

1. SOC Metrics

Key performance indicators include:

- MTTD (Mean Time to Detect): time taken to detect an incident



- MTTR (Mean Time to Respond): time taken to respond to an incident
- False Positive Rate: percentage of incorrect alerts
- Dwell Time: time attacker remains undetected

2. Executive Reporting

Executive reports simplify technical data into:

- summaries
- charts and dashboards
- risk-based insights
- recommendations for improvement

3. Continuous Improvement

Metrics help in:

- improving detection rules
- reducing response time
- enhancing SOC efficiency
- optimizing workflows

Practical Knowledge

Practical 1: Metrics Dashboard

Tool: Elastic Security

Example Metrics:

- MTTD = 2 hours
- MTTR = 4 hours
- False Positive Rate = 15%

Observation:

Dashboard provides real-time visibility of SOC performance.

Practical 2: Metrics Analysis



Tool: Google Sheets

Dwell Time Analysis:

- Attacker active before detection: 3 hours
- Investigation time: 2 hours

Observation:

Long dwell time indicates need for improved detection systems.

Practical 3: Executive Report

Summary (150 words):

The SOC performance analysis was conducted using key security metrics including MTTD, MTTR, false positive rate, and dwell time. The analysis showed that the average detection time was 2 hours and response time was 4 hours, indicating moderate efficiency in incident handling. However, dwell time analysis revealed that attackers remained undetected for up to 3 hours in some cases, highlighting gaps in detection capabilities. The false positive rate was measured at 15%, suggesting a need for better alert tuning. Based on these findings, it is recommended to improve automated detection rules, enhance threat intelligence integration, and optimize SOC workflows to reduce response times and improve accuracy. These improvements will strengthen overall SOC performance and reduce risk exposure.

Practical 4: Metrics Summary

Summary (50 words):

SOC performance was evaluated using key metrics such as MTTD, MTTR, false positive rate, and dwell time. Results showed moderate detection and response efficiency, with scope for improvement in alert accuracy and response speed. Recommendations include automation, better tuning of alerts, and improved threat intelligence integration.

Tools Used



- Elastic Security
- Google Sheets
- SOC dashboards
- SIEM tools
- Reporting tools

Skills Learned

- SOC performance measurement
 - executive reporting
 - metric analysis
 - data visualization
 - security communication
-

8. Capstone Project: Comprehensive SOC Incident Response

Introduction

The Capstone Project is a complete SOC workflow simulation that combines threat hunting, detection, automation, adversary emulation, incident response, forensic analysis, and reporting. It represents a real-world cybersecurity incident lifecycle handled by a Security Operations Center (SOC).

It is important for:

- understanding end-to-end SOC operations
- simulating real cyberattacks and response
- integrating multiple security tools
- improving detection and response skills
- practicing professional incident handling workflow

The main goal is to simulate a full cyberattack scenario and manage it from detection to reporting using SOC tools and frameworks.



Core Concepts

1. SOC Incident Lifecycle

- Attack Simulation
- Detection and Monitoring
- Triage and Analysis
- Response and Containment
- Automation (SOAR)
- Post-Incident Analysis
- Reporting

2. MITRE ATT&CK Mapping

Attacks are mapped to techniques such as:

- T1210 → Exploitation of Remote Services
- T1566 → Phishing
- T1078 → Valid Accounts

3. Incident Response Framework (NIST)

- Preparation
- Detection & Analysis
- Containment
- Eradication
- Recovery
- Lessons Learned

Practical Knowledge

Practical 1: Attack Simulation

Tool: Metasploit

Exploit Used:

exploit/multi/samba/usermap_script

Observation:



- vulnerability successfully exploited in target system
- unauthorized access achieved in controlled environment

Practical 2: Adversary Emulation

Tool: MITRE Caldera

Scenario:

Simulated T1210 attack (exploitation of remote services)

Observation:

- attack behavior replicated
- detection logs generated in Wazuh

Practical 3: Detection and Triage

Tool: Wazuh

Timestamp	Source IP	Alert Description	MITRE Technique
2025-08-18 16:00:00	192.168.1.102	Samba exploit	T1210

Observation:

SOC successfully detected exploit activity in real-time.

Practical 4: Response and Containment

Actions:

- isolate infected VM
- block attacker IP using CrowdSec
- verify blocking using ping test

Observation:

Containment successfully stopped further attacker activity.



Practical 5: SOAR Automation

Tool: TheHive

Automation Steps:

- detect alert
- validate with threat intelligence
- block malicious IP
- create incident ticket

Observation:

Automated workflow reduced manual response time.

Practical 6: Post-Incident Analysis

Method:

- 5 Whys RCA
- Fishbone Diagram

Observation:

Root cause identified as weak service security and lack of patching.

Practical 7: Metrics Reporting

Metrics:

- MTTD: 2 hours
- MTTR: 4 hours
- Dwell Time: 3 hours

Observation:

SOC performance is moderate with improvement required in detection speed.

Practical 8: Final Reporting



Executive Summary (150 words):

A comprehensive SOC incident response simulation was conducted involving a Samba service exploitation attack using Metasploit. The attack was successfully detected using Wazuh and mapped to MITRE ATT&CK technique T1210. Adversary emulation was performed using MITRE Caldera to validate detection capabilities. The incident was contained by isolating the affected system and blocking the attacker IP using CrowdSec. SOAR automation via TheHive improved response efficiency by automating alert validation and ticket creation. Post-incident analysis identified weak service security and lack of patch management as root causes. SOC performance metrics indicated moderate efficiency with an MTTD of 2 hours and MTTR of 4 hours. Recommendations include improving patch management, enhancing detection rules, and increasing automation coverage.

Tools Used

- Metasploit
- Wazuh
- CrowdSec
- TheHive
- MITRE Caldera
- Elastic Security
- Google Docs

Skills Learned

- full SOC lifecycle management
- incident detection and response
- adversary emulation
- SOAR automation
- forensic analysis
- executive reporting
